

Camera Chromaticity & Gamut Mapping Tool

Technical Overview & User Guide

Version 2 · CIE 1931 xy · Polar Plot · DNG ForwardMatrix 2

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Self-contained single-file HTML — no server, no dependencies

1. Overview

The Camera Chromaticity & Gamut Mapping Tool is a self-contained, single-file HTML application that visualises the colour gamuts of professional digital cameras and compares them against industry-standard target colour spaces. It requires no server, no installation, and no network connection — open the HTML file in any modern browser.

The tool is built on the CIE 1931 xy chromaticity diagram and presents two complementary plot types: a traditional CIE 1931 xy diagram and an xy polar plot centred on the working white point. Camera characteristics are derived from DNG ForwardMatrix 2 values extracted with ExifTool.

1.1 Key Capabilities

- Plot native sensor gamuts as triangles on the CIE 1931 xy diagram and polar plot
- Compare up to ten cameras simultaneously; multi-select up to two for detailed analysis
- Visualise gamut mapping to six target colour spaces using four distinct strategies
- Bradford chromatic adaptation between D50 and D65 working white points
- Quantitative analysis: gamut area %, per-primary hue and chroma deltas
- Natural-language comparative narrative for primary hue shift direction and magnitude

1.2 Technical Basis

Camera gamuts are defined by DNG ForwardMatrix 2 (FM2), which encodes the linear transform from balanced camera raw RGB to CIE XYZ under D65. Extracting the matrix columns yields the chromaticity coordinates of each primary. Chromatic adaptation uses the Bradford transform with ICC-standard matrix values, consistent with professional ICC colour management workflows.

A note on sub-locus primaries: modern camera sensors can produce primary chromaticities mathematically outside the spectral locus. These are valid virtual primaries — a property of sensor physics, not a data error.

2. Interface Layout

The interface is organised into three zones: the control bar at the top, the plot area in the centre, and the legend and analysis panel below. All controls update the plot in real time.

2.1 Control Bar — Row 1

Camera selection

All — activates all cameras. **None** — deactivates all. Individual camera buttons toggle each camera independently. The first click when **All** is active switches to that camera alone; subsequent clicks add/remove cameras, allowing any combination.

For analysis: select exactly one camera + target for camera-vs-space analysis; exactly two cameras (with or without a target) for camera-vs-camera comparison.

White Point

D50 (5003 K)	ISO standard for graphic arts and ICC profiling. Default selection.
D65 (6504 K)	Daylight standard; native WP for sRGB, Adobe RGB, DCI-P3, and all FM2 calibrations.

2.2 Control Bar — Row 2

Target Colour Space

Selects a reference gamut. When active, a dashed triangle shows the mapped gamut under the selected strategy. The analysis panel includes area % relative to the target.

Colour Space	Native WP	R xy	G xy	B xy	Notes
sRGB	D65	(0.640,0.330)	(0.300,0.600)	(0.150,0.060)	Web / screen
Adobe RGB (1998)	D65	(0.640,0.330)	(0.210,0.710)	(0.150,0.060)	Photography / print
DCI-P3/Display P3	D65	(0.680,0.320)	(0.265,0.690)	(0.150,0.060)	Cinema / displays
ProPhoto RGB	D50	(0.7347,0.2653)	(0.1596,0.8404)	(0.0366,0.0001)	Archival / raw
Hasselblad RGB	D50	(0.681,0.314)	(0.213,0.739)	(0.133,0.047)	Phocus output
Hasselblad Lstar	D50	(0.756,0.244)	(0.098,0.902)	(0.019,0.022)	Phocus Lstar

None — no target, only native gamuts drawn.

Show: Native

Toggles the solid-filled native camera gamut triangles. When off, only the mapped dashed triangles are drawn — useful for seeing mapped positions clearly.

View

CIE 1931	Standard horseshoe chromaticity diagram. x: 0.0–0.8, y: 0.0–0.9. Wavelength tick marks on locus. Default view.
Polar	Polar projection centred on the current white point. Radius = chroma distance from WP. Emphasises hue angle.

Mapping Strategy

Determines how the camera gamut is compressed or clipped to fit inside the target. Active only when a target colour space is selected.

Strategy	Description	In-gamut primaries	Shape behaviour
Hard Clip	Each primary moved along the WP ray to the exact target boundary. In-gamut primaries untouched.	Untouched	Distorts asymmetrically
Compress	Single scale factor applied to the whole triangle — just large enough to fit inside target.	Scaled equally	Preserved exactly
Hue-Preserve	Each primary moved to boundary at its exact hue angle from WP (binary search).	Untouched	Distorts by hue rays
Chroma Map	Proportional boundary-to-boundary remap: $r_{\text{mapped}} = r_{\text{src}} \times (r_{\text{tgt_bnd}} / r_{\text{src_bnd}})$.	Scaled proportionally	Matches target structure

3. Camera Referene

All cameras use DNG ForwardMatrix 2 (FM2), calibrated at D65. To add a camera, extract FM2 with ExifTool: **exiftool -ForwardMatrix2 yourfile.dng**

Enter the nine returned values as the **m** array in a **forwardMatrix([...])** call, set **nativeWP**: 'd65', choose a unique **color**, **label**, and **family** key, then add a matching button in the HTML.

Button	Camera	Format	FM2 WP
IQ4	Phase One IQ4	MF 100MP	D65
X2DII	Hasselblad X2D II	MF 100MP	D65
GFXII	Fujifilm GFX100 II	MF 102MP	D65
Z8	Nikon Z8	FF 45MP	D65
R5II	Canon R5 II	FF 45MP	D65
Q3	Leica Q3	FF 60MP	D65
M11	Leica M11	FF 60MP	D65
fpL	Sigma fp L	FF 61MP	D65
X-T5	Fujifilm X-T5	APS-C 40MP	D65
GRIV	Ricoh GR IVx	APS-C 26MP	D65

4. Reading the Plots

4.1 CIE 1931 xy Diagram

The curved boundary — the spectral locus — represents monochromatic light from ~380 nm to 700+ nm. A colour gamut is shown as a triangle whose corners are the R, G, B primary chromaticities. Solid fill = native sensor gamut; dashed outline = gamut after mapping. The central dot marks the active white point. The four reference space triangles (sRGB, Adobe RGB, DCI-P3, ProPhoto) are drawn as faint dashed outlines.

TIP Hover over the CIE 1931 diagram to read exact xy chromaticity coordinates in the status bar.

4.2 Polar Plot

The polar plot re-projects CIE 1931 data with the white point at centre. Radial distance = chroma (saturation); angular position = hue. Makes hue angle differences and chroma extension easy to compare.

4.3 Outside the Spectral Locus

Some sensor primaries appear outside the spectral locus. This is physically real: silicon photodetectors have broad spectral sensitivities producing virtual primaries beyond monochromatic limits. They are a property of the sensor matrix, not a data error.

5. Analysis Panel

The analysis panel appears automatically beneath the legend when exactly one or exactly two cameras are selected.

5.1 Single Camera vs Target Colour Space

Activate one camera and select a target colour space. Two blocks appear side-by-side.

Left block: Gamut Area

Bar chart showing: Native gamut % of target area; Target at 100%; Mapped % after applying the selected strategy. Narrative explains whether the camera exceeds or falls short of the target.

NOTE	Native > 100% means the camera captures colours outside the target space. Native < 100% means some target colours cannot be natively captured.
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Right block: Primary Hue Shift

Three rows (R, G, B) compare the camera primary to the target space primary:

Δhue°	Hue angle difference from white point. Positive = clockwise/warmer; negative = counter-clockwise/cooler.
$\Delta\text{chroma} \%$	Chroma distance from WP, thousandths of xy unit. Positive = more saturated.
$\Delta x \%$	Raw x chromaticity difference, thousandths of xy unit.
$\Delta y \%$	Raw y chromaticity difference, thousandths of xy unit.
Area %	Triangle area (shoelace formula) as percentage of the target or reference gamut.

Colour-coded tags: Warm (orange) = shifted toward orange/yellow; Cool (blue) = toward purple/cyan; Green = B shifted toward cyan; Neutral = change below significance threshold.

5.2 Two-Camera Comparison

Activate exactly two cameras. A target is optional.

The area block shows both cameras as % bars normalised to the larger gamut. With a target, a second group shows both mapped areas as % of the target for a three-way comparison.

Two primary comparison blocks appear: Camera A vs B, and B vs A, with the same per-primary Δhue , Δchroma , Δx , Δy metrics and tags, giving a fully bidirectional picture.

TIP	Two cameras + a target simultaneously shows camera-vs-camera comparison with mapped area bars for both against the target.
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6. Colour Science Background

6.1 CIE 1931 Chromaticity

The CIE 1931 colour matching functions describe how the human visual system responds to light across the visible spectrum. Dividing tristimulus values XYZ by $X+Y+Z$ yields chromaticity coordinates xy, which represent colour direction independent of luminance.

6.2 Gamuts and Colour Spaces

A colour space defines a set of primaries and a white point. The gamut is the triangle formed by connecting the R, G, B primary chromaticities. Triangle area in xy space is a useful — if approximate — measure of gamut size. It is proportional to the range of encodable colours, though it ignores the luminance dimension.

6.3 DNG ForwardMatrix

DNG ForwardMatrix (cam→XYZ) column vectors directly encode the XYZ tristimulus values of each camera channel, giving the most direct route to sensor primary chromaticities. FM2 is paired with CalibrationIlluminant2 (D65 for all cameras here). The nine values are row-normalised so neutral maps to $Y=1$.

6.4 Bradford Chromatic Adaptation

The Bradford CAT converts tristimulus values between illuminants by scaling in a cone-like LMS space. This tool uses ICC-standard Bradford matrix values for $D65 \leftrightarrow D50$ conversion, consistent with ICC colour management.

6.5 Gamut Mapping in Practice

All four strategies operate in chromaticity space (xy) rather than perceptual space. This makes the mathematics transparent and visually inspectable:

- **Hard Clip** — closest point on the boundary along the WP ray. Most direct mechanical approach.
- **Compress** — uniform saturation reduction. Simplest approximation of photographic tonal range compression.
- **Hue-Preserve** — hue is the last attribute sacrificed; used in many ICC rendering intent implementations.
- **Chroma Map** — proportional gamut-boundary-to-boundary remap. Preserves relative chroma position — colours at 60% of native gamut saturation land at 60% of target saturation. In-gamut primaries scale down proportionally. Closest xy-space analogue to ICC perceptual rendering intent.

Key difference between Compress and Chroma Map: Compress treats the triangle as a rigid body to be shrunk uniformly; Chroma Map treats the gamut as a rubber sheet stretched proportionally between the native and target boundaries.

7. Workflow Examples

7.1 Camera vs sRGB for Web Delivery

1. Click None, then select one camera (e.g. Q3).
2. Select sRGB as Target Colour Space.
3. Read the Area block: >100% means the camera captures colours outside sRGB.
4. Read the Primary Shift block: a "shifted cyan/cooler" G tag means the camera green is more cyan than sRGB, affecting skin tone and foliage rendering after conversion.
5. Cycle through all four strategies and observe how the mapped triangle shape and Mapped % change.

7.2 Comparing Two Medium-Format Cameras

6. Click None, then click IQ4 and X2DII.
7. The analysis panel appears without a target.
8. Read the Area block to see which sensor has the larger gamut.
9. Read both primary comparison blocks. R primary Δ hue: positive = warmer red, negative = cooler/more purple.
10. Select HasselbladRGB as the target to see how each camera maps into that output space.

7.3 Understanding White Point Effects

11. Select any single camera and target.
12. Note the primary coordinates under D50.
13. Switch to D65. The reference dot moves and primary coordinates update via Bradford adaptation.
14. Δ hue and Δ chroma values remain accurate regardless of WP selection.

7.4 All Cameras Overview

15. Click All, select CIE 1931 view, set Target to None.
16. Observe spread of gamut triangles — medium format tends to show wider gamuts, especially in the green channel.
17. Add a target (e.g. ProPhoto) to see all native gamuts against the reference.

7.5 Comparing All Four Mapping Strategies

Select one wide-gamut camera (e.g. IQ4) and sRGB as the target.

18. **Hard Clip**: each primary pinned to sRGB boundary along the WP ray. G typically moves further than R and B, distorting the shape substantially.
19. **Compress**: whole triangle shrinks uniformly. Shape is preserved. If native gamut is much larger, the mapped triangle will be very small because the worst primary determines the scale factor.

20. **Hue-Preserve:** each primary moved to boundary along its hue angle. Shape differs from both Clip and Compress. Hue-ray and WP-ray diverge for off-axis primaries.
21. **Chroma Map:** proportional remap. In-gamut primaries also move — scaled proportionally down. More even distribution than Hard Clip or Compress. In the Polar plot, each mapped primary lies at exactly the same angle as the native primary.

8. Technical Reference

8.1 Data Pipeline

22. Extract: `exiftool -ForwardMatrix2 file.dng`
23. Enter nine values into `forwardMatrix([...])` call in the HTML source
24. Normalise: scale by $1 / (m[3] + m[4] + m[5])$ so neutral maps to $Y=1$
25. Extract primaries: feed $[1,0,0]$, $[0,1,0]$, $[0,0,1]$ through the normalised matrix; convert XYZ to xy
26. Adapt: Bradford D65→D50 (or identity) to display white point
27. Plot, then apply gamut mapping if a target is selected

8.2 Bradford Matrix Values

D65 to D50 (ICC specification):

```
[ 1.0478112, 0.0228866, -0.0501270 ]
[ 0.0295424, 0.9904844, -0.0170491 ]
[-0.0092345, 0.0150436, 0.7521316 ]
```

D50 to D65:

```
[ 0.9555766, -0.0230393, 0.0631636 ]
[-0.0282897, 1.0099416, 0.0210077 ]
[ 0.0122982, -0.0204830, 1.3299098 ]
```

8.3 Gamut Area Calculation

Shoelace formula applied to the three primary xy coordinates:

Area = $| (Rx(Gy-By) + Gx(By-Ry) + Bx(Ry-Gy)) | / 2$

Expressed as % relative to the target gamut area.

8.4 Chroma Map Algorithm

For each primary vertex:

28. $\theta = \text{atan2}(\text{primary.y} - \text{WP.y}, \text{primary.x} - \text{WP.x})$
29. $r_src = \sqrt{((\text{primary.x} - \text{WP.x})^2 + (\text{primary.y} - \text{WP.y})^2)}$
30. **r_src_bnd** = radius at which the ray from WP at angle θ crosses the native gamut boundary
31. **r_tgt_bnd** = radius at which the same ray crosses the target boundary
32. **r_mapped** = $r_src \times (r_tgt_bnd / r_src_bnd)$
33. **Mapped primary** = $\{ \text{WP.x} + \cos(\theta) \cdot r_mapped, \text{WP.y} + \sin(\theta) \cdot r_mapped \}$

Boundary radius is found by ray–edge intersection against each triangle edge. A primary at 80% of the source boundary maps to 80% of the target boundary. An out-of-gamut primary (ratio > 1.0) maps beyond the target and is clamped.

8.5 Strategy Comparison

Hard Clip	In-gamut untouched · Shape distorts asymmetrically · Best for: exact per-primary boundary positions
Compress	All primaries scaled equally · Shape preserved exactly · Best for: uniform saturation reduction
Hue-Preserve	In-gamut untouched · Shape distorts by hue rays · Best for: preserving perceived hue
Chroma Map	In-gamut scaled proportionally · Shape matches target structure · Best for: relative chroma preservation (ICC perceptual analogue)

8.6 Adding New Cameras

Locate the `presetMatrices` object in the HTML source and add an entry:

```
"MyCam_FM2": {
  m: forwardMatrix([r1c1,r1c2,r1c3, r2c1,r2c2,r2c3, r3c1,r3c2,r3c3]),
  nativeWP: 'd65', color: '#RRGGBB', label: 'Camera Name', family: 'mycam'
},
```

Also add a button to the Camera row in the HTML controls with the same `data-preset` value as family.

9. Troubleshooting

Analysis panel not showing	Select exactly 1 camera + target, or exactly 2 cameras. Panel hides for all other selections.
Mapped triangle very small	Expected in Compress mode when native gamut vastly exceeds the target. Switch to Hard Clip or Chroma Map to see per-primary results with less crushing of in-gamut primaries.
Chroma Map ≈ Hue-Preserve	Both preserve hue angle. Chroma Map also scales in-gamut primaries proportionally; Hue-Preserve leaves them at their native position.
Primary outside horseshoe	Physically real for sensor primaries — silicon photodetectors produce virtual primaries beyond monochromatic limits. Not an error.
Black camera label invisible	Hasselblad X2D II uses #000000. Change the color field in presetMatrices to a lighter hex value.
All/None stuck	These set the entire selectedCameras Set at once. Reload the page to reset state if needed.
Polar crossing lines	The polar plot clips triangles to the circular boundary. Extreme primaries appear at the rim; connecting edges can arc near the boundary.

10. Quick Reference Card

Controls

All / None	Select all cameras / deselect all
Camera button	Toggle camera (first click when All is on → switch to that camera only)
D50 / D65	White point: affects polar centre, Bradford adaptation, analysis coordinates
Target: None	Show native gamuts only, no mapping
Target: sRGB etc.	Show native + mapped (dashed); enable single-camera analysis
Show: Native	Toggle solid-filled native triangle on/off
CIE 1931	Horseshoe diagram, x 0–0.8, y 0–0.9
Polar	xy polar projection centred on active white point
Hard Clip	Per-primary clip along WP ray to boundary; in-gamut untouched
Compress	Uniform scale of whole triangle to just fit target; shape preserved
Hue-Preserve	Per-primary move to boundary at constant hue angle from WP
Chroma Map	$r_mapped = r_src \times (r_tgt_bnd / r_src_bnd)$; proportional, perceptual

Analysis triggers

1 camera + any target	Area % comparison + per-primary hue/chroma shift vs target
2 cameras, no target	Area % comparison + bidirectional per-primary analysis
2 cameras + target	Area % with both mapped areas + bidirectional primary comparison
Any other selection	Analysis panel hidden

Analysis metrics

Area %	Triangle area (shoelace) as % of target or larger camera gamut
Δhue°	+ve = clockwise/warmer; –ve = counter-clockwise/cooler
$\Delta chroma\%$	+ve = more saturated (further from WP); –ve = less saturated
$\Delta x\%$	Raw x chromaticity difference, thousandths
$\Delta y\%$	Raw y chromaticity difference, thousandths